



Fundamental study on anodizing of aluminum and aluminum alloys in a sodium tetraborate electrolyte solution and evaluating their high-temperature wettability behavior



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Highlights

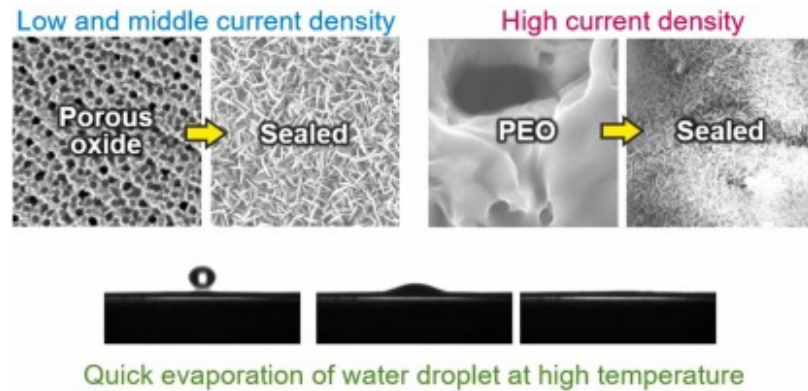
- Porous alumina with feather-like inner pore structures was formed by anodizing in a Na₂B₄O₇-based electrolyte.
- Plasma electrolytic oxidation (PEO) films were generated under high current density conditions.

- Both porous alumina and PEO films were successfully hydrated by boiling water sealing, enhancing surface hydrophilicity.
- Contact angle tests showed increasing surface hydrophilicity in the order: etched/desmutted surface > porous alumina > PEO.
- Droplet tests at 423–473 K showed anodized oxides suppress Leidenfrost effect and promote rapid evaporation.

Abstract

We investigated the anodizing process of aluminum (Al) and Al alloys using a sodium tetraborate electrolyte solution. This study focused on the formation of porous alumina and evaluated its wettability behavior at high temperatures. High-purity Al and several Al alloys (A1050, A3004, A5052, and A7075) were anodized under varying current densities to examine the structural characteristics of the oxide films. The results demonstrated that sodium tetraborate-based anodizing enabled the formation of porous alumina with feather-like structures on the inner pore walls. In addition, anodizing at high current densities formed plasma electrolytic oxidation (PEO) films. A subsequent pore-sealing treatment facilitated the hydration of alumina, irrespective of whether it was a porous alumina or PEO film. We further evaluated the interaction between water droplets and heated anodized surfaces, providing insights into their applicability in high-temperature environments. These findings contribute to the advancement of Al surface treatments for improved thermal management and industrial applications.

Graphical Abstract



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Keywords

Aluminum; Anodizing; Porous alumina; Plasma electrolytic oxidation; Leidenfrost effect; Evaporation

1. Introduction

Anodizing is a crucial process used to enhance the corrosion resistance and mechanical properties of the surface of aluminum (Al) and its alloys, which is applied in nanotechnology and other fields [1], [2], [3], [4], [5]. As various structures can be fabricated by varying the aqueous electrolyte solution used in anodizing, the chemical species of electrolytes were explored [6], [7], [8], [9], [10], [11], [12], [13], [14]. Studies have mainly focused on acidic aqueous solutions. However, in recent years, anodizing processes using alkaline electrolytes have attracted attention. Anodizing Al in sodium metaborate or sodium phosphate solution formed

porous alumina with a thick barrier layer at the bottom [15], [16]. When ammonium carbonate was used, a structure combining typical porous oxide and plasma electrolytic oxidation (PEO) films was formed [17].

Sodium tetraborate is also an alkaline electrolyte that has been studied for several decades [18], [19], [20] and has been extensively pursued in recent years. The use of sodium tetraborate leads to the formation of highly ordered porous alumina with feather-like bumps on the inner walls of pores by controlling the anodizing conditions [21], [22]. In addition, porous alumina formed using sodium tetraborate exhibits higher corrosion resistance than porous alumina formed by anodizing using other typical acidic electrolytes [23]. Note that the focus of this study is on anodizing using sodium alkaline tetraborate solution and the resulting porous alumina, which differs from the smooth barrier-type oxide film produced by anodizing using boric acid-sodium tetraborate buffer solutions, as already extensively reported [24], [25]. The industrial application of anodizing using alkaline electrolytes, especially sodium tetraborate, requires the investigation of the resulting nanostructures formed on pure Al and practical Al alloys. Moreover, structural changes after the pore-sealing process need to be confirmed, as this considerably affects the final performance of the anodized film [26], [27], [28], [29].

The Leidenfrost effect is a unique phenomenon related to wettability in high-temperature environments. A liquid droplet dropped on the surface of a hot substrate evaporates instantly, causing a vapor film to form at the interface between the substrate and the liquid droplet. Hence, the liquid droplet floats [30], [31], [32], [33]. Al materials are commonly used in high-temperature (>373 K) or water-cooled environments in the engineering field. Therefore, evaluating the behavior of water droplets when dropped onto porous alumina heated to > 373 K is an important indicator for using porous alumina in high-temperature environments. Herein, we evaluate nanostructures formed by anodizing pure Al and practical Al alloys immersed in aqueous solutions of alkaline sodium tetraborate ($\text{Na}_2\text{B}_4\text{O}_7$). We observe the behavior of water droplets upon dropping them onto the heated surfaces of these anodized specimens.

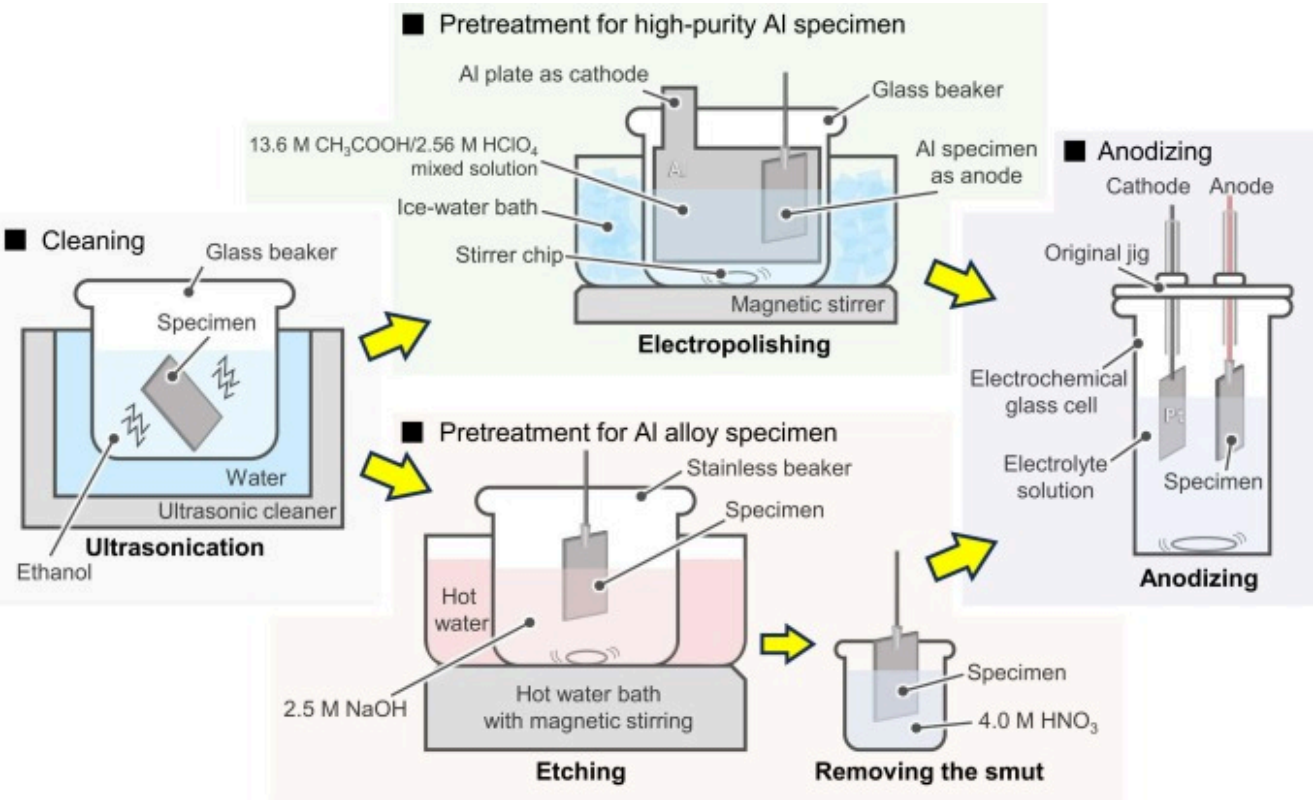
2. Experimental

High-purity Al sheets (99.999 mass%) and Al alloy sheets (A1050, A3004, A5052, and A7075) were used as specimens. The compositions of the alloys are presented in Table 1. Each Al sheet was cut to have a reaction area of 10 mm wide and 20 mm long connected to a 5 mm × 15 mm conductive handle. After cutting, the specimens were immersed in ethanol and ultrasonically cleaned for 10 min. The high-purity Al specimens were further immersed in a mixture of 13.6 M CH_3COOH /2.56 M HClO_4 mixed solution at 280 K and electropolished at a constant voltage of 28 V for 2 min. The alloy specimens were immersed in an aqueous

2.5 M NaOH aqueous solution at 333 K for 2 min for etching and then immersed in a 4.0 M HNO₃ aqueous solution at room temperature for 30 s to remove the smut. After each process, the specimens were washed using ultrapure water (Milli-Q water, 18.2 MΩ·cm). The volume of the electrolyte solution was 150 ml, and it was stirred at 250 rpm during anodizing. The pretreated specimens were immersed in a 0.3 M Na₂B₄O₇ solution at 343 K, and constant-current anodizing was performed at a current density (*i*) of 80–240 A m⁻² for 30 min. A 99.95 mass% Previous studies have examined anodizing at various concentrations of sodium tetraborate [21]. Since 0.3 M is the concentration that most reliably forms an anodized film, the concentration of the aqueous solution was determined to be 0.3 M. In preliminary experiments, uniform porous alumina was not formed at electrolyte temperatures below 333 K. Therefore, the anodizing temperature was fixed at 343 K to ensure stable oxide film growth. The reason for choosing the constant-current anodizing method is that it is more widely used industrially than the constant-voltage method. The applied current density was adjusted based on the following reasons: when the value was too low, oxide growth was retarded and chemical dissolution predominated, while excessively high current densities led to local burning. Pt sheet was used as a cathode. The size of the Pt electrode plate is approximately 400 mm². The anode and cathode are placed parallel to each other, with a distance of approximately 20 mm. A water bath (UCT-1000A, AS ONE, Japan) was used to control the temperature of the solution. A direct current (DC) stabilized power supply (PWR400H, KIKUSUI, Japan) was used to apply the current, and a digital multimeter (VOAC7602, IWATSU, Japan) was used to record the voltage values. The anodized specimens were immersed in boiling water for 15 min to seal the pores. These specimen preparation processes are shown in Fig. 1. The nano-microstructures of the surfaces and fracture surfaces of these specimens were observed via field emission scanning electron microscopy (FE-SEM, JSM-6500F, JEOL, Japan) and atomic force microscopy (AFM, Nanocute, Hitachi High-Technologies, Japan). FE-SEM observations were conducted using specimens coated with Pt to prevent charging, with an accelerating voltage of 10 kV. Secondary electrons were detected for the observations. AFM images were acquired at multiple locations, and representative images are presented. The cross-sections of films formed on various alloys were observed via FE-electron probe microanalysis (FE-EPMA, JXA-8530F, JEOL, Japan). The accelerating voltage was set to 10 kV. Specimens were coated with carbon to prevent charging prior to FE-EPMA analysis. The contact angle of the specimen surface was measured using a contact angle meter (SM-CE1, Kyowa Interface Science, Japan) using Milli-Q water as the liquid. Water droplets with a volume of 4 μL were used during the contact angle measurements. The crystallinity of the specimen surface was evaluated using XRD (SmartLab, Rigaku, Japan). The diffraction peaks were indexed based on the standard reference data (PDF 1512488, 1010461, 1010412, ICDD).

Table 1. Chemical composition of four aluminum alloys (mass%) used in this study.

Alloy	Element (mass%)									
	Fe	Si	Mg	Cu	Mn	Mg	Zn	Ti	V	Al
A1050	0.32	0.11	-	0.02	-	-	-	0.02	0.01	bal.
A3004	0.7	0.3	0.8–1.3	0.25	1.0–1.5	0.8–1.3	0.25			bal.
A5052	0.24	0.09	2.57	0.01	-	2.57	0.01	0.18	0.03	bal.
A7075	0.34	0.09	2.69	1.4	0.03	2.69	5.47	0.08	0.2	bal.



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Fig. 1. Schematic illustration of the specimen preparation process, excluding the cleaning step with ultrapure water.

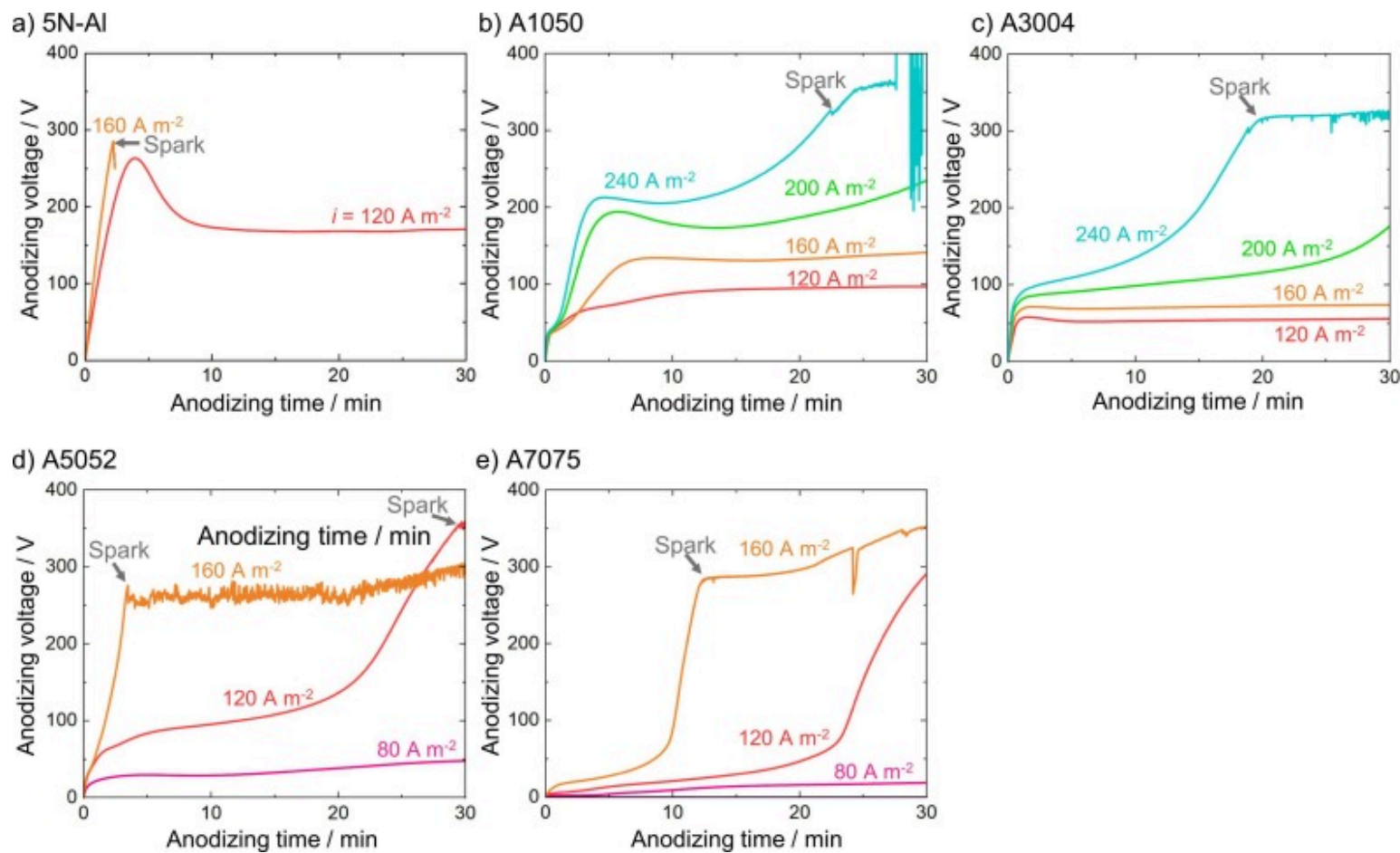
To evaluate the behavior of water droplets dropped onto a heated porous alumina surface, a custom experimental setup was constructed and used for the evaluation, as described below. The reaction area of the specimen was enlarged to 20 mm × 20 mm, and anodizing was performed under the abovementioned conditions. Specimens were heated from the rear using a hot plate heater (C-MAG HP 7, IKA Japan, Japan). To avoid the effects of temperature oscillations caused by the heater being switched on and off to maintain the hot plate temperature, a high-purity copper plate measuring 10 mm in width, 20 mm in length, and 5 mm in thickness was placed between the specimen and hot plate. A syringe pump (SPE-1, AS ONE, Japan) was used to drop water droplets (Milli-Q water). During this process, temperature changes on the anodized specimen were measured using a sheet-type K-type thermocouple and data logger (GL260, Graphtec, Japan). The thermocouple was fixed to the specimen surface using heat-resistant Kapton tape. A high-speed camera recorded at 1000 fps (MHU130EX-K, Shodensha, Japan) was used to record a video of water droplets falling onto the surface of a specimen. The tests were conducted in a controlled environment maintained at a temperature of 298 K and a humidity of 55 % using a precision air conditioner (PAP mini PAP01B-KJ, ORION MACHINERY, Japan) connected to a large globe box (800 × 600 × 620 mm, AS-800S-L, AS ONE, Japan).

3. Results and discussion

3.1. Anodizing of high-purity Al and Al alloys using a Na₂B₄O₇ solution

Fig. 2a shows the voltage–time curves of 5N-Al sheets immersed in a 0.3-M Na₂B₄O₇ solution at 343 K and anodized at constant i for 30 min. At an i of 120 A m⁻², the voltage increases linearly at the beginning of anodizing, then drops slightly, and later becomes steady. This change in voltage is a typical behavior observed during anodizing using acidic electrolytes such as sulfuric acid, oxalic acid, phosphoric acid, and etidronic acid [34], [35], [36], [37]. As i increases to 160 A m⁻², the voltage sharply increases during the initial stage of anodizing, then the voltage increase becomes impossible, and spark discharge is observed at that moment. As a result, a uniform oxide film could not be formed. The arrows in the graph indicate the starting point of spark discharge, which continued thereafter. The arrows in the following figure have the same meaning. Fig. 2b shows the results of the anodizing process of A1050. At a low i of 120 A m⁻², no voltage peak appears at the beginning of anodizing. However, when $i > 160$ A m⁻², voltage peaks (135 and 194 V) appear at the beginning of anodizing. At an i of 240 A m⁻², the voltage change during the initial stage of anodizing is similar to that observed at a low current density. However, after 22 min, spark discharge occurs as the

voltage increases, and a uniform film is not formed after 30 min anodizing. The anodizing voltages of A3004 (Fig. 2c) at i values of 120 and 160 A m⁻² exhibit a steady value after the initial transient period, while the voltage increases gradually at an i of 200 A m⁻² after the initial transient period and does not exhibit a steady value. Upon applying a high i of 240 A m⁻², the voltage increases after the initial stage and spark discharge occurs on the specimen surface at around 300 V, making the voltage relatively constant. The voltage of A5052 (Fig. 2d) is constant at an i of 80 A m⁻². At 120 A m⁻², the voltage behavior A5052 is similar to those of A1050 and A3004 at 200–240 A m⁻², with a small spark when the voltage reaches around 350 V immediately before the anodizing time of 30 min. Meanwhile, after the initial anodizing voltage increases linearly at an i of 160 A m⁻², spark discharge occurs over the entire specimen at about 260 V and anodizing continues with voltage oscillations of approximately 260–300 V. This behavior is similar to the PEO film, which is also observed when other acidic electrolytes are used [38], [39]. The anodizing behavior of A7075 (Fig. 2e) is similar to that of A5052. However, when the same current density is applied to A7075 and A5052, the resulting voltage of A7075 is lower. Spark discharge occurs from the entire surface of the specimen of A7075 at approximately 300 V at an applied current density of 160 A m⁻², whereas no significant voltage oscillation occurs at this time. Moreover, at the same current density, the resulting voltage is lower for Al alloys than for high-purity Al.



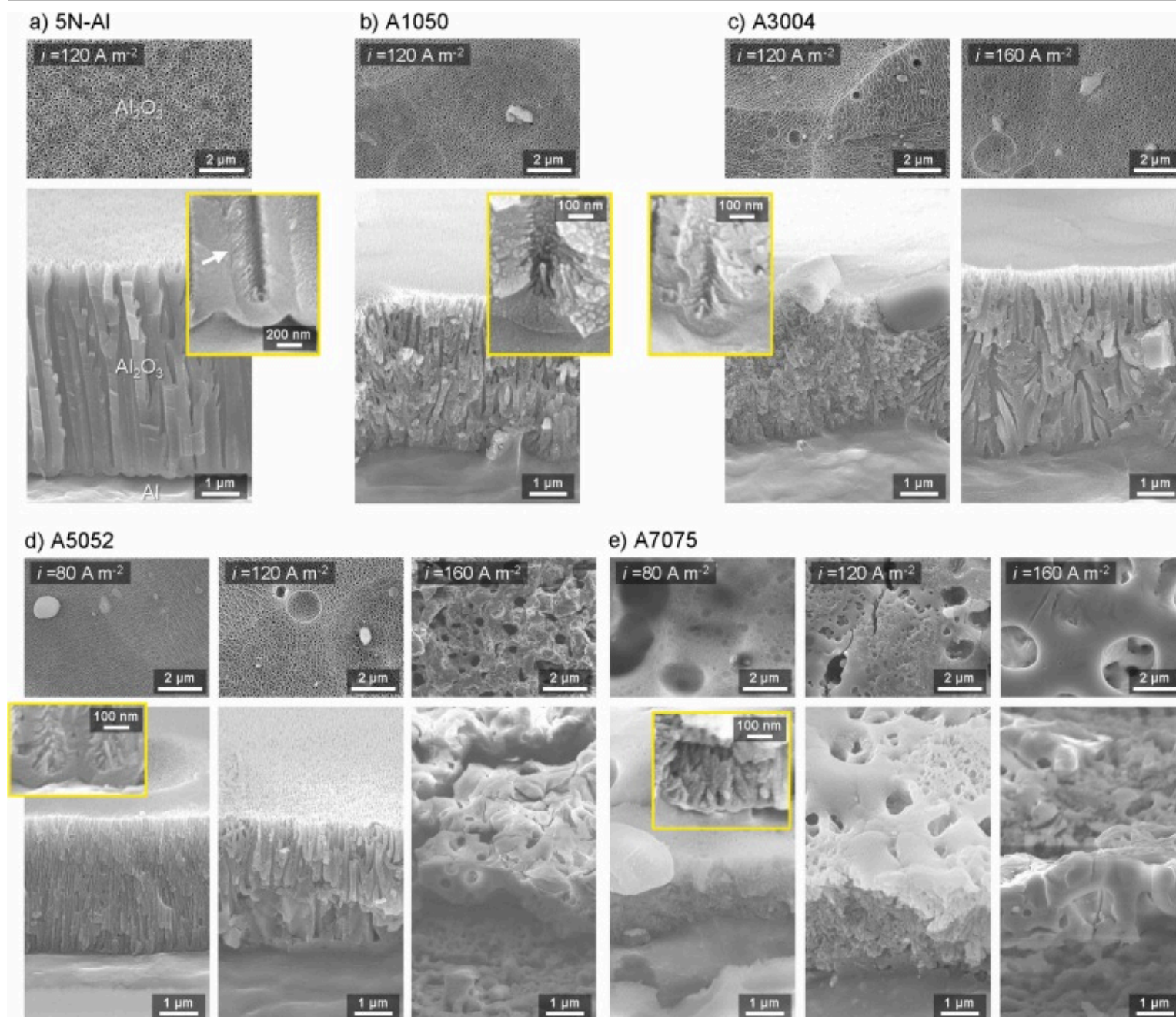
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Fig. 2. Changes in the anodizing voltage as a function of the anodizing time for a) 5N-Al, b) A1050, c) A3004, d) A5052, and e) A7075 specimens in a 0.3-M $\text{Na}_2\text{B}_4\text{O}_7$ solution at a T of 343 K.

Fig. 3 shows the SEM images of the surface and fractured cross-sectional surface of porous alumina obtained via anodizing (Fig. 2). The yellow square shows a magnified image of the bottom part of the film. Pores with extremely random arrangement are present on 5N-Al anodized in a 0.3 M- $\text{Na}_2\text{B}_4\text{O}_7$ solution at 343 K at a i of 120 A m^{-2} . The cross-section of this specimen shows that while there are several small pores in the upper part of the oxide layer (the oxide layer formed in the early stages of anodizing),

as the anodizing time increases, the pores are selected and branched pores are formed. The inner walls of the pores have a structure with several small, feather-like bumps, which is common when anodizing is performed using an alkaline electrolyte solution [16], [21], [22]. Fig. 3b shows porous alumina fabricated on A1050 after anodizing at 120 A m^{-2} . The specimen surface exhibits a wavy pattern of around $5 \mu\text{m}$, which is ascribed to etching, and an intermetallic compound of $\sim 1 \mu\text{m}$. Similar to anodizing using high-purity Al, pore walls of an anodized A1050 specimen show branched porous alumina with feather-like structures on the inner wall of the pores. A3004 anodized at an i of 120 A m^{-2} exhibits wavy patterns, intermetallic compound lumps, microscopic pores, and feather-like structures on pore walls, similar to those of A1050. The fracture surface shows an oxide film formed with intermetallic compounds wrapped around it. A high i of 160 A m^{-2} results in a high voltage, and the barrier layer and pore walls in porous alumina become thick. The thickness of the oxide film also increases. An oxide film of A5052 similar to that of other examined alloys is obtained at $80\text{--}120 \text{ A m}^{-2}$. Meanwhile, as i increases to 160 A m^{-2} , large pores of approximately $0.5\text{--}1 \mu\text{m}$ in pore diameters appear on the surface, and a structure significantly different from those of other films is observed. This film lacks vertical pores and exhibits a sponge-like structure, suggesting the formation of a PEO film. A7075 results in a surface with large dimples, wavy patterns of $\sim 2 \mu\text{m}$, and small dimples ($\sim 200 \text{ nm}$) at an i of 80 A m^{-2} . The cross-section of the film reveals that the film thickness is slightly uneven. Although the pore structure is small because of the low voltage during anodizing, a porous structure comprising a hemispherical barrier layer and pore walls is observed under high magnification. It is well-known that the changes in the nanostructures occur before and after the appearance of voltage peaks during the formation of a porous structure [40]. However, even when voltage peaks are not observed in the current–time curve (Fig. 2), a well-defined porous structure is formed. This suggests that the formation of porous alumina can occur regardless of the presence or absence of voltage peaks. At higher i of 160 A m^{-2} , as in the case of the A5052 alloy, the formation of PEO is observed by high i anodizing of A7075 alloy. As described above, the use of sodium tetraborate for anodizing enables the formation of either porous or PEO-type oxide films by controlling the current density.



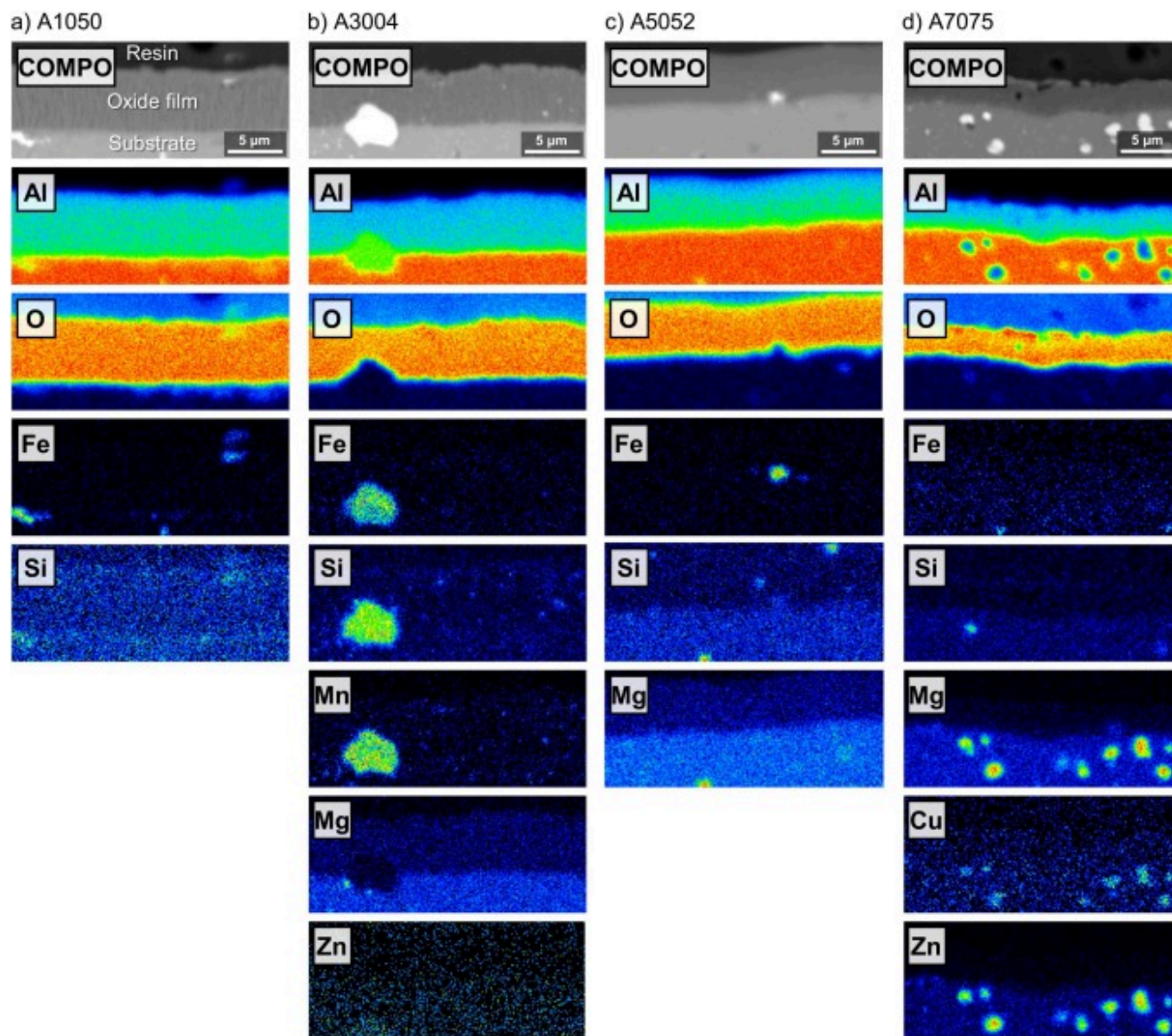
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Fig. 3. Surface and cross-sectional surface SEM images of a) 5N-Al, b) A1050, c) A3004, d) A5052, and e) A7075 specimens anodized in a 0.3-M $\text{Na}_2\text{B}_4\text{O}_7$ solution at a T of 343 K and i of 80–160 A m^{-2} .

A comparison of the cross-sections of porous alumina formed via anodizing at an i of 120 A m^{-2} reveals that the oxide film obtained using Al alloys is thinner than that obtained using high-purity Al, even considering the slight tilt observed on the fractured surface. Therefore, the current efficiency is expected to be lower when Al alloys are anodized than when high-purity Al is anodized. In principle, the anodic conversion efficiency of fabricating porous alumina cannot be rigorously determined, because oxide growth and chemical/electrochemical dissolution occur simultaneously [41]. This intrinsic competition makes a simple Faradaic calculation of film thickness unreliable. The situation becomes even more complex when commercial Al alloys are anodized. In principle, the anodic conversion efficiency of fabricating porous alumina cannot be rigorously determined, because oxide growth and chemical/electrochemical dissolution occur simultaneously. This intrinsic competition makes a simple Faradaic calculation of film thickness unreliable. The situation becomes even more complex when commercial Al alloys are anodized. The presence of intermetallic particle phases can promote local galvanic effects [42]. These heterogeneities disrupt the uniform growth of the oxide and facilitate localized dissolution, thereby decreasing the effective current efficiency. Consequently, anodic films formed on alloys tend to be thinner than those on high-purity Al under identical anodizing conditions.

Fig. 4 shows EPMA-based compositional images and the corresponding elemental mapping of the cross-sections of anodized oxides produced on various Al alloy substrates. The anodizing conditions are the same as those shown in Fig. 2. For ensuring the conditions for the fabrication of a porous oxide film, an i of 120 A m^{-2} was chosen for A1050, A3004, and A5052, while a low i of 80 A m^{-2} was chosen for A7075. Alloys containing iron (Fe) and silicon (Si) are observed in the substrate and oxide layer, indicating that an oxide film is formed, while Fe and Si are incorporated into the oxide film during anodizing. Although alloys contain magnesium (Mg), copper (Cu), and zinc (Zn) in the base metal, they are barely detected in the oxide layer, suggesting that these added elements including intermetallic compounds dissolve during anodizing. Large voids observed in the oxide film are ascribed to the dissolution of these elements, including intermetallic compounds [43].



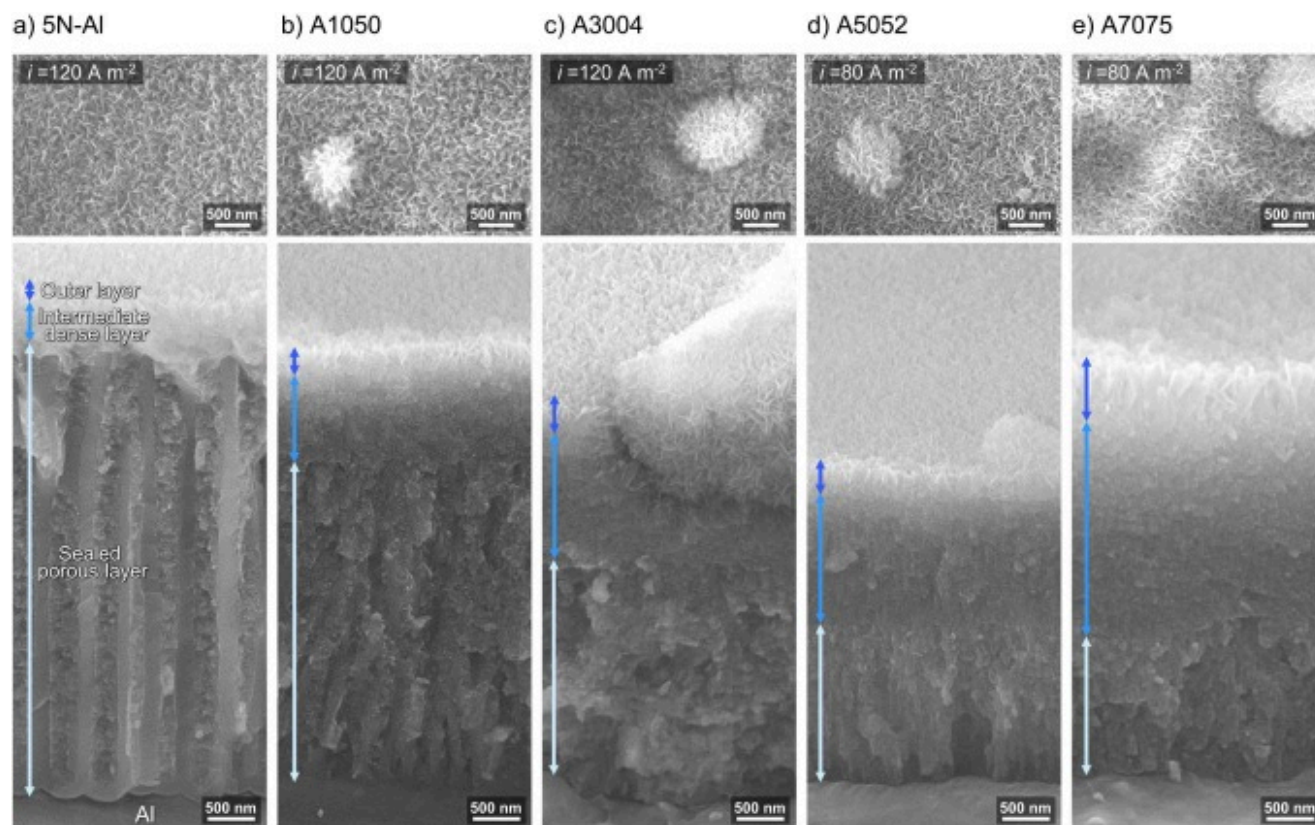
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Fig. 4. Cross-sectional SEM images and corresponding elemental distribution maps of a) A1050, b) A3004, c) A5052, and d) A7075 specimens anodized in a 0.3-M $\text{Na}_2\text{B}_4\text{O}_7$ solution at a T of 343 K and i of 120 A m^{-2} .

3.2. Pore-sealing behavior of the anodized oxide film fabricated using an aqueous $\text{Na}_2\text{B}_4\text{O}_7$ solution

In this study, pore-sealing treatment was applied primarily to confirm that porous alumina prepared in sodium tetraborate electrolyte can undergo effective sealing, since this process does not always proceed in other electrolytes such as phosphoric acid [44], [45]. In addition, the formation of hydrated alumina phases during sealing was expected to modify the surface hydrophilicity. Therefore, the sealing experiments were included to examine their influence on wettability and cooling behavior, rather than to assess corrosion resistance. Fig. 5 shows the SEM images of the surfaces and fractured cross-sectional surfaces of specimens anodized under the conditions shown in Fig. 2 (an i of 120 A m^{-2} for 5N-Al, A1050, and A3004 and 80 A m^{-2} for A5052 and A7075). Following anodization, the pores were sealed by immersing them in boiling water for 15 min. High-purity Al exhibits a three-layered structure: an (i) outer layer of rough, flaky pseudo-boehmite, (ii) middle layer of relatively dense pseudo-boehmite, and (iii) layer of hydrated Al oxide flakes inside the porous layer. The pores were sealed similar to those of porous alumina prepared using a typical acidic aqueous solution [45], [46]. A similar three-layered structure is observed for Al alloys. The results of the surface and cross-sectional areas of all alloys exhibit white contrast clusters, i.e., areas where intermetallic compounds are exposed and covered with flaky hydrated Al oxides. The oxides formed on the surface of intermetallic compounds might also form hydrates. Fig. 6 shows the surface structure of A5052 and A7075 anodized at an i of 160 A m^{-2} followed by immersing in boiling water for 15 min. The yellow square shows a magnified image of the specimen surface. The formation of numerous flakes on the surface of both specimens indicates that PEO films might be hydrated. Notably, the pore size of the PEO film is considerably larger than that of porous alumina, indicating that the pores are not truly sealed.

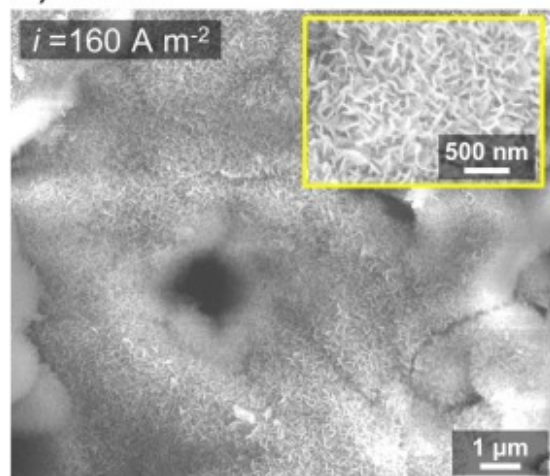


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Fig. 5. Surface and cross-sectional surface SEM images of a) 5N-Al, b) A1050, c) A3004, d) A5052, and e) A7075 specimens anodized in a 0.3-M $\text{Na}_2\text{B}_4\text{O}_7$ solution at a T of 343 K and i of 120 A m^{-2} followed by pore sealing.

a) A5052



b) A7075

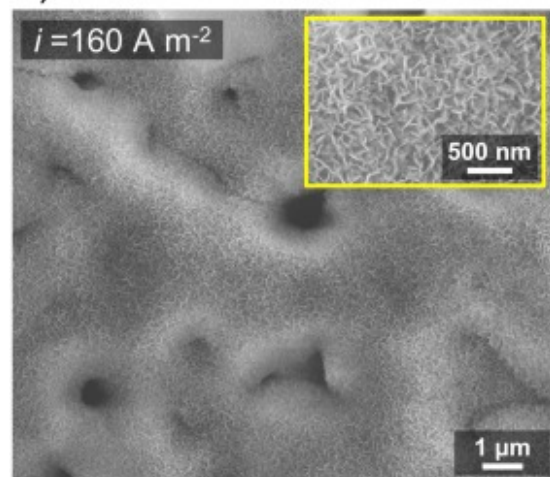
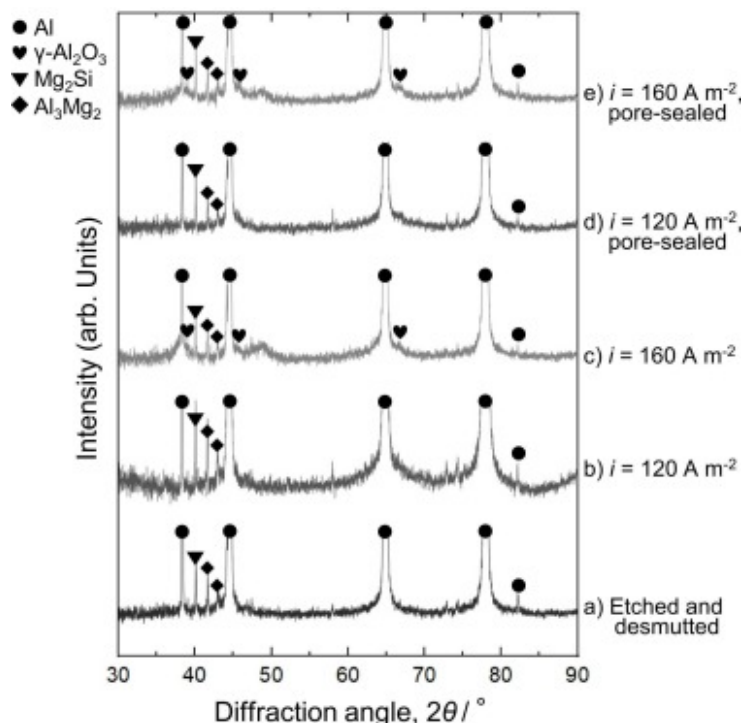
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Fig. 6. Surface and cross-sectional surface SEM images of the a) A5052 and b) A7075 specimens anodized in a 0.3-M $\text{Na}_2\text{B}_4\text{O}_7$ solution at a T of 343 K and i of 160 A m^{-2} followed by pore sealing.

The composition and crystallinity of the specimens were evaluated using XRD (Fig. 7). Spectrum a is obtained by XRD measurement of the etched and desmutted A5052 substrate. Peaks derived from base material of Al, Mg_2Si , and Al_3Mg_2 , which is known as an intermetallic compound of the A5052 alloy [47], were observed. Peaks at $2\theta = 58.0, 72.8, \text{ and } 74.3^\circ$, which are similarly detected in all specimens, are also considered to originate from the alloy matrix. As only a natural oxide film has formed on the substrate, the background is also relatively small. Spectrum b is from a specimen that has only porous alumina formed by anodizing at 120 A m^{-2} . The high background and the lack of clear peaks other than those of the substrate correspond to the amorphous structure of the porous alumina. Spectrum c shows the results of measuring the PEO film formed by anodizing at 160 A m^{-2} . The small peaks derived from $\gamma\text{-Al}_2\text{O}_3$ observed in addition to the substrate spectrum indicate the formation of the PEO film [48]. However, the peaks are not very clear because the crystallinity of the oxide film is low. The spectra in d and e are the results of measurements of the porous alumina and the PEO film after the pore-sealing treatment. Although a pseudo-boehmite layer is expected to form as a result of the pore-sealing treatment [49], [50], its extremely thin thickness made it difficult to detect any distinct peaks in the XRD patterns and reference patterns considered (e.g., PDF 1000061) did not match the observed diffraction peaks. The mechanical properties of oxide films, such as hardness and modulus, have been reported in numerous studies [2], [51], [52] and are strongly dependent on electrolyte type and anodizing conditions. Since the present work focuses on wettability and cooling behavior, mechanical testing was not carried out; however, the reported values for similar anodized films suggest that the films formed in sodium tetraborate fall within typical ranges.



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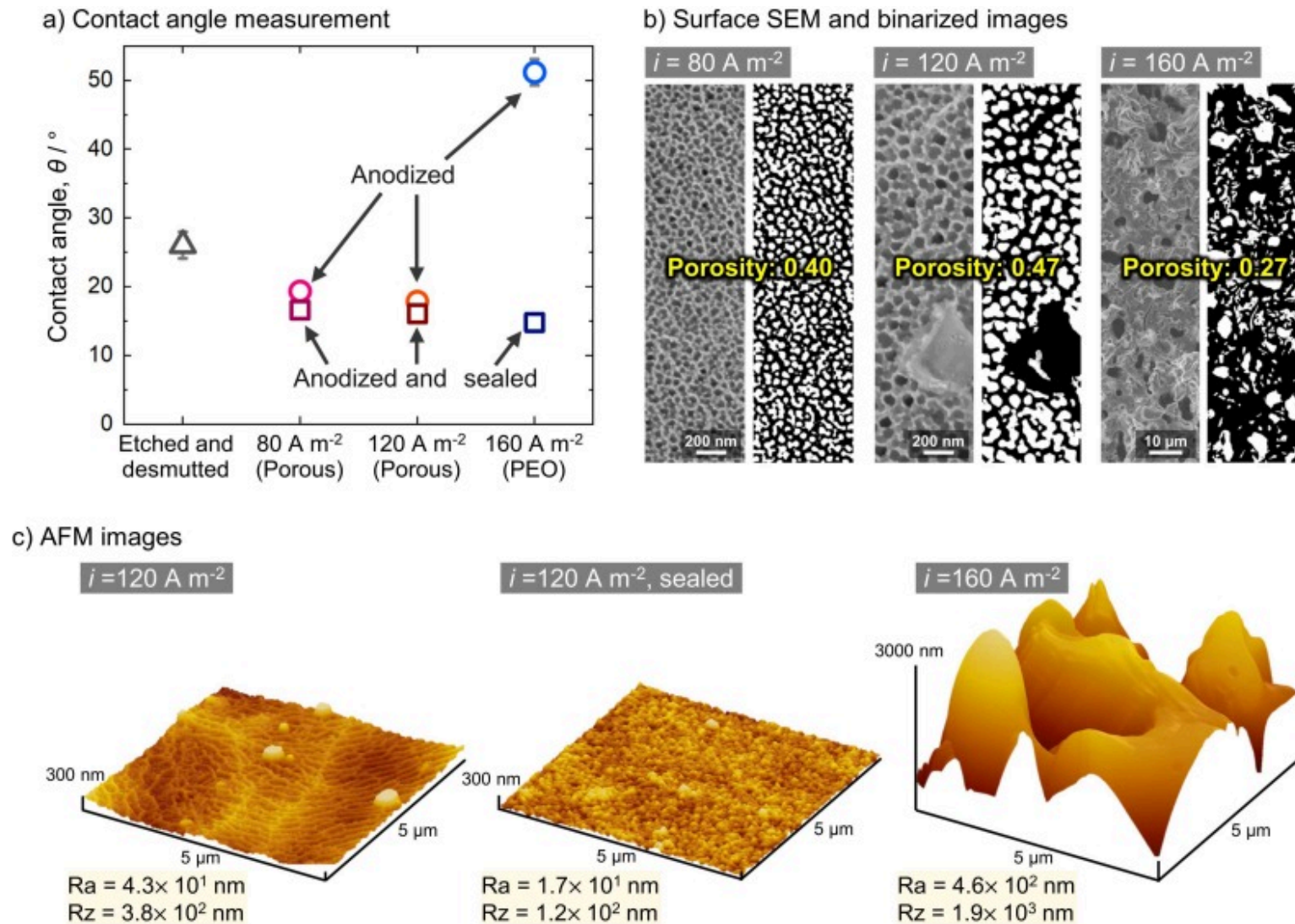
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Fig. 7. X-ray diffraction patterns of the A5052 aluminum: a) after etching and desmutting and b)–e) after the formation of porous alumina and PEO films, before and after pore sealing.

3.3. Behavior of liquid droplets dropped onto a hot substrate

Among Al alloys, A5052 is widely available on the market and is used in automotive components and building materials [53], [54], [55]. In cases of industrial applications, it is often used after forming porous alumina on the surface of the metal. Therefore, it is important to analyze the water cooling behavior when porous alumina and PEO film are formed on such materials. As discussed so far, anodizing in a sodium tetraborate solution enables controlled fabrication of both porous alumina and PEO films using A5052 alloy. Thus the Leidenfrost effect of porous alumina and PEO film fabricated on A5052 was evaluated by heating at high temperatures. The contact angle of a specimen surface is known to have a significant effect on heat transfer characteristics [56]. Fig. 8 shows the contact angle of A5052 with only pretreated (etched and desmuted), and porous alumina and PEO films

before and after pore sealing A5052 is relatively hydrophilic, and its hydrophilicity is further enhanced by the formation of an oxide film. However, the PEO films exhibited a contact angle of 51° , indicating moderate hydrophilicity. After pore sealing, it is substantially more hydrophilic owing to changes in the nanostructure and chemical properties. To clarify the reasons behind these changes, the surface structures were evaluated. Fig. 8b shows surface SEM images of A5052 anodized at an i of $80\text{--}160\text{ A m}^{-2}$, along with their binarized versions. The area fraction occupied by dark regions in the SEM images, which correspond to the pores, was measured to determine porosity. The porosity values at i of 80 and 120 A m^{-2} were 0.40 and 0.47, respectively, showing only slight difference. In contrast, a significant decrease to 0.27 was observed at i of 160 A m^{-2} . The pore diameters were also found to increase with increasing current density. Fig. 8c shows the surface structures of the porous alumina formed at i of 120 A m^{-2} , pore sealed porous alumina, and PEO films formed at 160 A m^{-2} measured by AFM. In the figure, R_a denotes the average surface roughness, and R_z denotes the maximum height roughness. Comparison of the surfaces before and after pore sealing indicates that the unsealed specimen exhibits irregularities along the pore walls and exposure of intermetallic particles. After the sealing treatment, the surface shows fine nanoscale roughness attributed to hydrated Al oxide flakes, and the influence of intermetallic compounds is less apparent. Consequently, both R_a and R_z decrease following sealing. R_a and R_z of the PEO film are more than ten times higher than those of the porous alumina, indicating the formation of considerably larger surface roughness. Altogether, these results suggest that the porous alumina formed at $80\text{--}120\text{ A m}^{-2}$ is characterized by high porosity, small pore diameter, and a dense distribution of pores, resulting in a Wenzel state in which water readily infiltrates the structure. In contrast, the PEO film formed at 160 A m^{-2} exhibits low porosity and a coarser, more complex surface morphology, which likely results in a state closer to the Cassie–Baxter state, where liquid infiltration is hindered. The enhanced hydrophilicity observed after the sealing treatment is attributed to the formation of hydrated Al oxide flakes, which introduce fine surface roughness and abundant hydroxyl ($-\text{OH}$) groups.

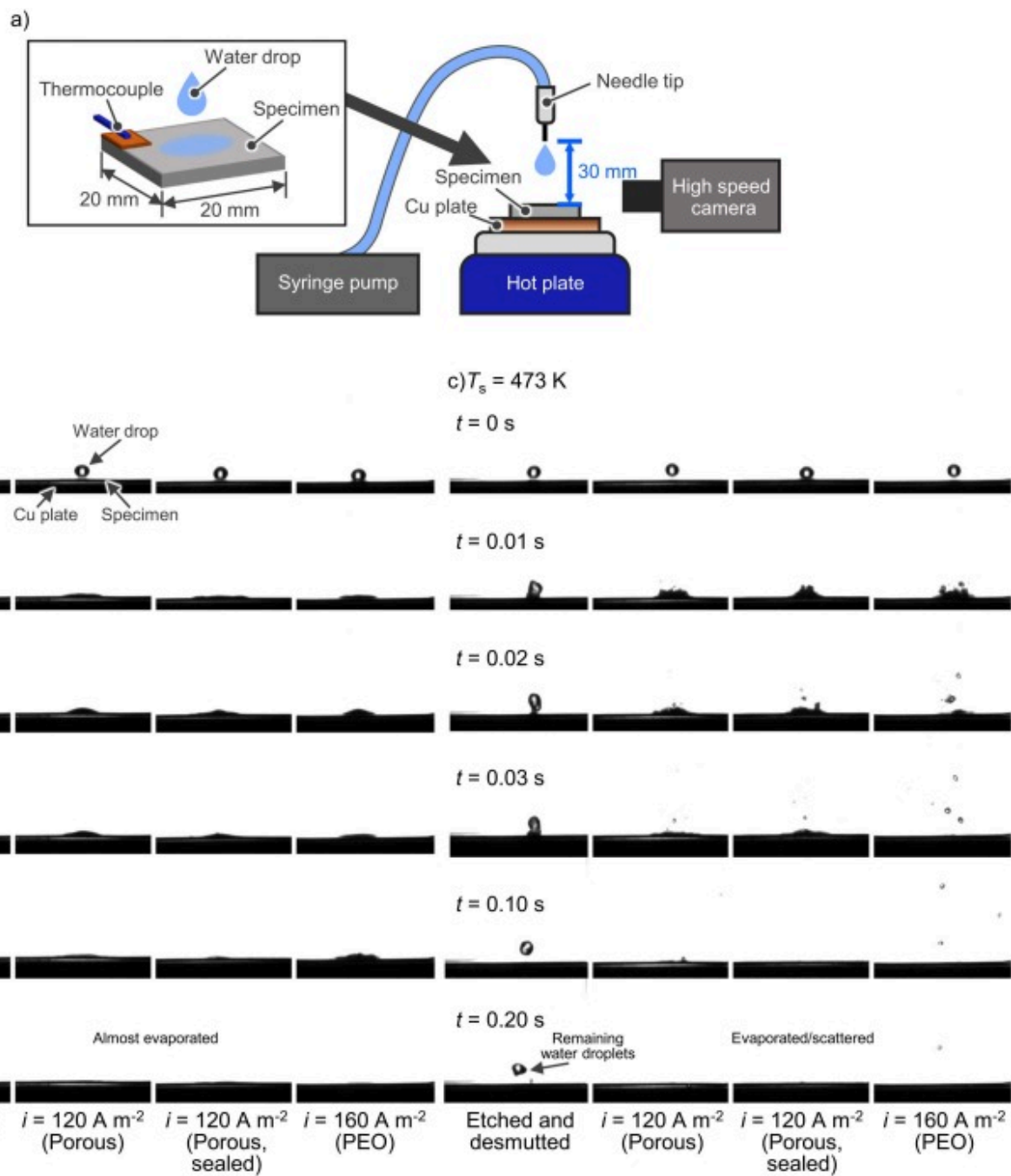


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Fig. 8. a) Contact angles, b) surface SEM and binarized images, and c) AFM images of the A5052 surface: after etching and desmutting and after the formation of porous alumina and PEO films before and after pore sealing.

The behavior of the droplets of these specimens mounted in the apparatus (Fig. 9a) during dripping was analyzed. Fig. 9b shows snapshots (captured using a high-speed camera) of samples with (i) only pretreated (etched and desmutted), and (ii) porous alumina, (iii) sealed porous alumina and (iv) PEO films heated to a T_s (surface temperature) of 423 K and then exposed to 6 μL of Milli-Q water dropped from a height of 30 mm (Supplementary Video 1). In each case, water droplets were observed to spread and gradually vaporize. Fig. 9c shows the results of a similar experiment where the substrate temperature increases to a T_s of 473 K (Supplementary Video 2). The pretreated specimen exhibits no spreading of water droplets, which remain as drops and floated on the substrate surface. This indicates the presence of the Leidenfrost effect. Meanwhile, most water droplets spread across the porous alumina and PEO surfaces of A5052, while some droplets break and scatter. Therefore, the structure of the specimen surface affects the behavior of water droplets.

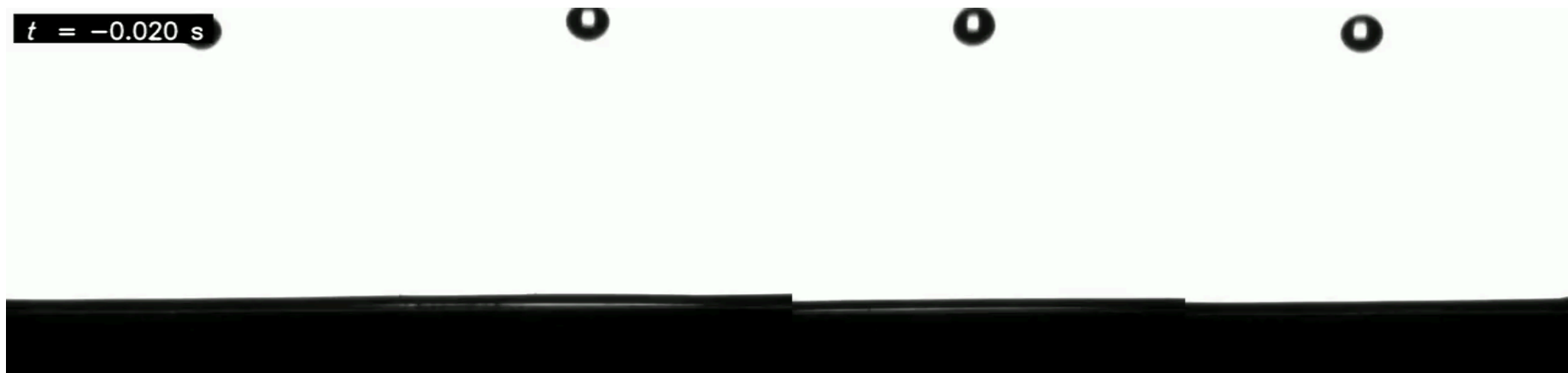


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Fig. 9. a) Apparatus used for the water drop test. b) and c) Snapshots from high-speed video showing liquid droplets deposited onto various substrate surfaces at T_s of b) 423 and c) 473 K.

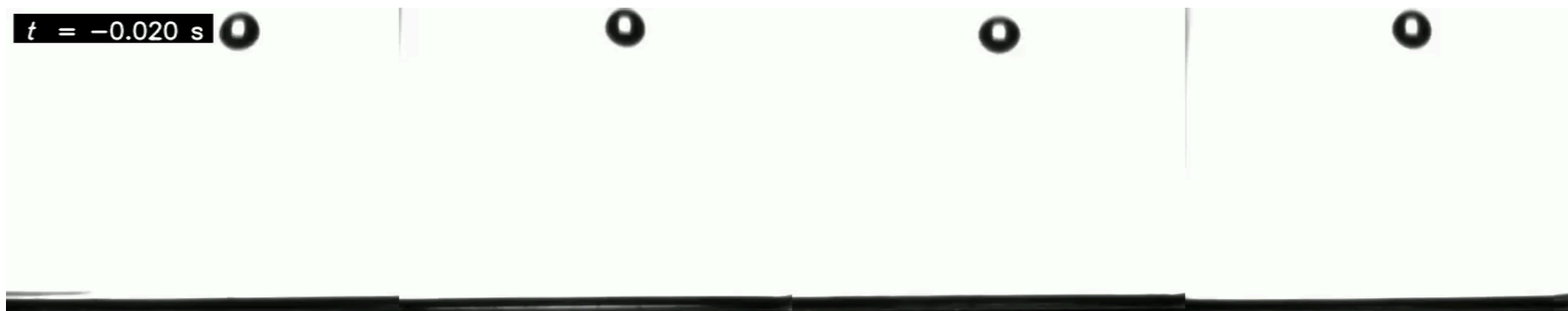
The following is the Supplementary material related to this article [Video S1](#).



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Video S1 Liquid droplets deposited onto various substrate surfaces at $T_s = 423$ K. From left: Etched/desmutted, 120 A m^{-2} (Porous), 120 A m^{-2} (Porous + Sealed), 160 A m^{-2} (PEO)..

The following is the Supplementary material related to this article [Video S2](#).

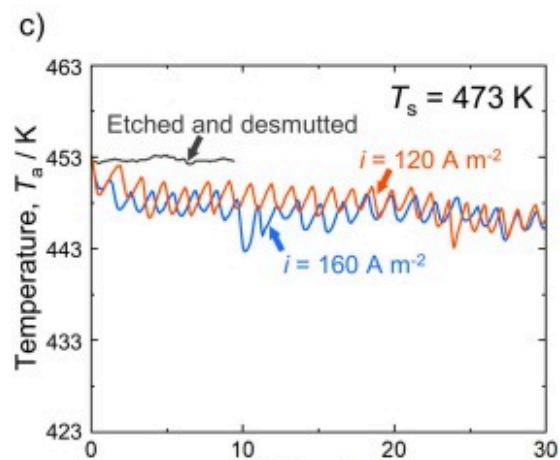
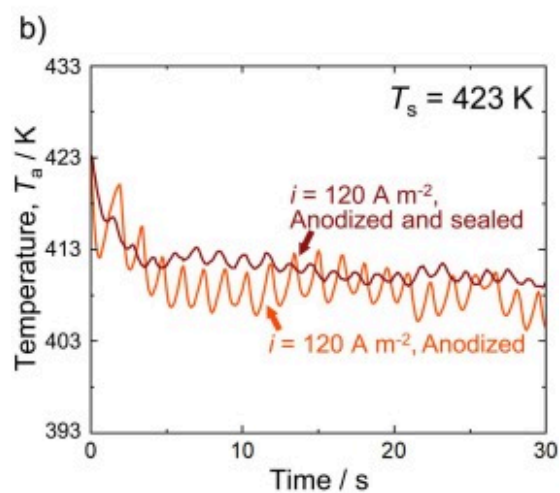
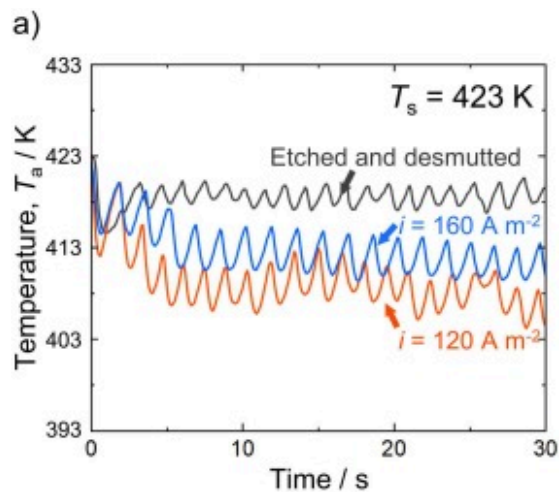


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Video S2 Liquid droplets deposited onto various substrate surfaces at $T_s = 473$ K. From left: Etched/desmutted, 120 A m^{-2} (Porous), 120 A m^{-2} (Porous + Sealed), 160 A m^{-2} (PEO)..

Fig. 10a shows changes in the temperature of the specimen surface (T_a) measured using a thermocouple during a continuous water drop test (Fig. 9b). The specimen was continuously heated from the rear side using a heater via a Cu plate. Although the specimen surface temperature decreases with each drop of water for the pretreated samples, T_a remains nearly unchanged even after water droplets are continuously added. Alternately, T_a significantly decreases with each drop of water for the specimens with porous alumina and PEO films formed via anodizing at i values of 120 and 160 A m^{-2} , which gradually decreases later until it reaches a steady state. The temperature was 420 K for the pretreated specimen, 404 K for the porous alumina film, and 410 K for the PEO film 30 s after water droplets began to fall. The difference in surface temperature reduction, i.e., the difference in cooling efficiency, is presumed to result from the water droplets spreading less easily on the PEO film than on the porous alumina, as indicated by the contact angle measurements. Fig. 10b shows changes in T_a during a continuous water drop test on porous alumina anodized at an i of 120 A m^{-2} before and after pore sealing. The temperature drop caused by the initial droplet is high for unsealed porous alumina. As additional droplets are added, the surface temperature of the unsealed specimen continues to decrease more than that of the sealed specimen. After 30 s, the surface temperature of the sealed specimen reaches 413 K. Based on these results, it is supposed that the thermal conductivity of hydrated alumina is lower than that of alumina. The hydrated alumina layer formed via the pore-sealing treatment inhibits heat transfer, slowing the evaporation of water droplets. This suggests that the rate of water evaporation per unit time is reduced on the specimen with the hydrated alumina layer. Several studies have been studied on the thermal conductivity of porous alumina and PEO films [57], [58], [59].; however, due to the complex interaction of multiple substances and structural parameters, such as porous structure and thin flake-like structure, it is difficult to discuss specific values for thermal conductivity. Fig. 10c shows the results of a water drop test conducted using the same specimen as Fig. 10a with the substrate temperature increased to 473 K. Fig. 9c shows that water droplets jump and roll from the pretreated specimen surface without spreading. Consequently, the surface temperature of the pretreated specimen does not decrease. The temperature measurement of the pretreated specimen was discontinued because water droplets rolled off and adhered to the thermocouple, complicating accurate temperature monitoring. In contrast, water droplets spread across the surface and evaporated on the porous alumina and PEO films, decreasing surface temperature. However, compared with the condition at a low initial temperature, a higher number of water droplets are scattered on contact with the surface, suggesting that the temperature decrease per droplet is smaller at high substrate temperatures. Therefore, the cooling efficiency of the substrate after water spraying on a high-temperature substrate surface is higher for porous alumina and PEO films than that for

the pretreated Al specimen. The cooling efficiency of the typical porous alumina film is higher when the substrate temperature is relatively low. The difference in cooling efficiency between porous alumina and PEO films decreases when the substrate temperature is high, and the cooling efficiency decreases when a sealing treatment is performed. These results suggest that the film structure and material properties have a significant effect on the cooling efficiency of these specimens. Conventional fabrication processes for surfaces designed to enhance heat-transfer efficiency often involve expensive fabrication equipment or complicated multi-step processes [31], [32], [33]. In contrast, anodizing offers a low-cost and scalable approach that can be readily applied to large surface areas, which is particularly important for practical thermal management applications.



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Fig. 10. Changes in T_a with time during the water drop test: a) etched and desmuted surface vs. anodized surface at 423 K, b) anodized surface vs. sealed surface at 423 K, and c) etched and desmuted surface vs. anodized surface at 473 K.

4. Conclusions

Anodizing high-purity Al and Al alloys in an alkaline $\text{Na}_2\text{B}_4\text{O}_7$ solution resulted in the formation of porous alumina and PEO films at low and high i values, respectively. Both films could be sealed by immersing in boiling water. The contact angles of the surfaces decreased in the mentioned order: etched and desmuted surface > porous alumina film surface > PEO film surface. This indicated increasing hydrophilicity of the surfaces. Sealing further enhanced the hydrophilic nature of the surfaces. Water droplet experiments in the range of 423–473 K revealed that etched and desmuted surfaces exhibited the Leidenfrost effect at 473 K, resulting in prolonged droplet evaporation times. In contrast, droplets on porous oxide and PEO films rapidly evaporated at the same temperature. These results demonstrated the influence of the structure of the oxide surface and wettability on high-temperature droplet behavior.

CRedit authorship contribution statement

Hiroki Habazaki: Writing – review & editing. **Mana Iwai:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Tomoya Nagao:** Writing – review & editing, Investigation. **Sho Kitano:** Writing – review & editing. **Koji Fushimi:** Writing – review & editing. **Takahiro Noguchi:** Writing – review & editing, Investigation. **Tatsuya Kikuchi:** Writing – review & editing, Methodology.

Declaration of Competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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